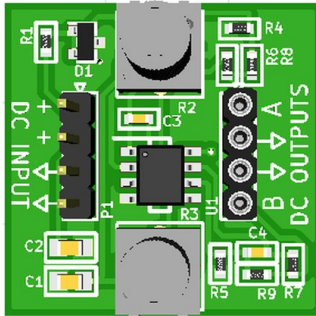


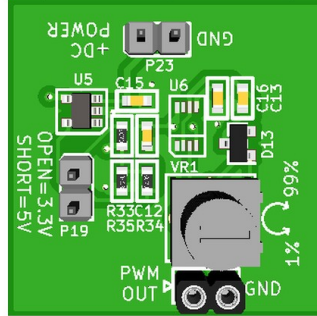
LED Driver Experimenter's Kit

User Manual and Laboratory Guide

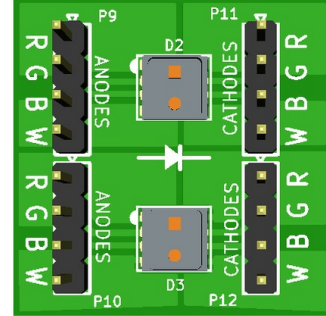
Nine reusable LED circuits on one V-scored panel



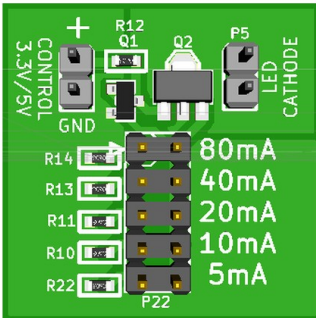
1 Analog source



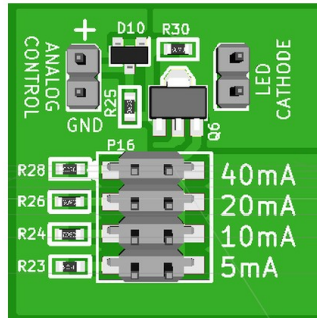
2 PWM source



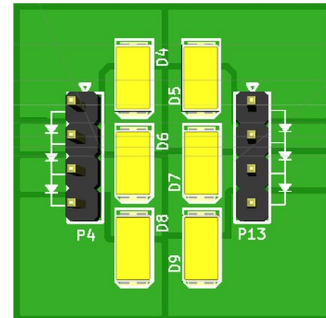
3 RGBW LEDs



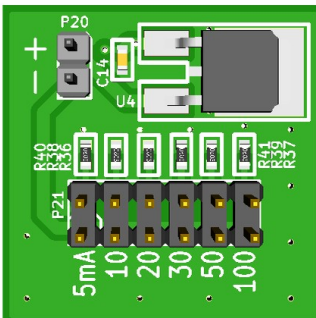
4 PWM linear driver



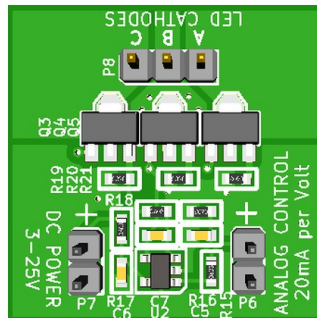
5 Analog linear driver



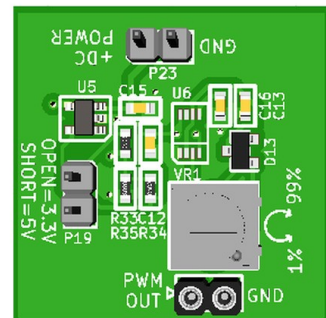
6 White LEDs



7 Series driver



8 Triple driver



9 Switch-mode driver

Revision A • September 2026

Dangermouse Industries LLC

Welcome to the Kit

This kit is a compact laboratory for learning how LEDs are driven and dimmed. The nine circuits include analog and PWM control sources, two LED-load boards, several linear constant-current topologies, and a high-efficiency switching driver. Use the panel intact while learning, or snap the V-scored panel into nine 1 inch by 1 inch modules for prototypes and permanent projects.

Begin with the board-reference sections to understand each module and its limits. Then use the application setup diagrams at the end of the manual to assemble complete working combinations.

What You Need

- A current-limited DC bench supply. A 12 V to 18 V supply is suitable for many of the application setups.
- Insulated jumper wires or miniature test leads suitable for the board connectors.
- A digital multimeter. An oscilloscope is optional but especially useful for PWM experiments.
- Eye protection and a nonconductive work surface.

How to Use the V Scored Panel

The circuits arrive as one panel. Keeping the panel intact makes the board numbers and connections easy to follow. If you separate the modules, support the panel close to a score line and bend only along that line. Separate one row or column at a time. Lightly smooth sharp fiberglass edges after separation and remove all debris before powering a circuit.

Important: Snapping a panel creates sharp fiberglass edges and dust. Wear eye protection, keep dust away from electronics, and clean the work area afterward.

Safety

Bright-light hazard: The LED boards can produce uncomfortable or potentially harmful light levels. Do not look directly into illuminated LEDs. Begin every experiment at the lowest practical current.

- Turn off the power supply before moving wires or changing jumpers.
- Use the bench supply current limit. Begin with a conservative limit and increase it only after checking the wiring.
- Observe polarity. Reverse polarity can damage semiconductors and LEDs.
- Some linear drivers and LEDs become hot. Disconnect power and allow them to cool before touching them.
- Never exceed an individual board's stated voltage, current, or control-input limit.
- The kit is an open-frame experimental product. Keep conductive objects away while power is applied.

Board Overview

Board	Function	Use
1	Dual analog source	Two adjustable 0 V to 5 V control outputs
2	PWM generator	Adjustable duty cycle; 3.3 V or 5 V output
3	Dual RGBW LEDs	Two four-color LED loads
4	PWM linear driver	Low-side current sink; 5 mA to 155 mA
5	Analog linear driver	Voltage-controlled low-side sink; 5 mA to 75 mA
6	White LED strings	Two strings of three high-brightness LEDs
7	Series linear driver	Two-terminal high- or low-side regulator; 5 mA to 215 mA
8	Triple analog driver	Three matched outputs; approximately 20 mA/V
9	Switch-mode driver	Efficient buck driver; up to 350 mA

Basic Concepts

LEDs Need Current Regulation

LED forward voltage changes with device type, current, and temperature. A small voltage change can cause a large current change, so LEDs should not be connected directly across a voltage source. Each driver in this kit controls current in a different way.

Linear and Switch Mode Drivers

A linear driver dissipates the unused voltage as heat. Its approximate dissipation is (supply voltage minus total LED voltage) multiplied by LED current. Linear circuits are simple and useful for studying current regulation, but they require thermal care. Board 9 transfers energy through an inductor and switching device, which generally reduces dissipation when the voltage difference or current is large.

Analog and PWM Dimming

Analog dimming changes the regulated LED current continuously. PWM dimming keeps the on-state current fixed and changes the fraction of time the LED is on. Board 4 follows PWM directly. Board 8 filters the PWM waveform and converts its average value into a DC current command.

Current Setting

On boards with multiple current jumpers, enabled values add. For example, enabling 100 mA and 50 mA produces a nominal 150 mA setting. Always change jumpers with power removed.

Common Ground

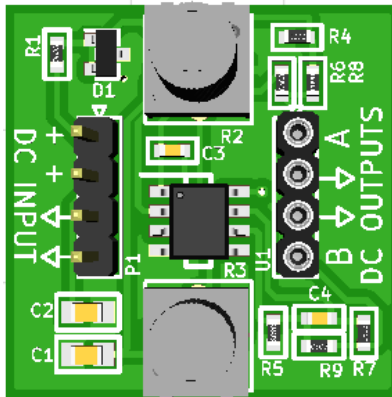
Control sources and controlled drivers normally require a shared ground reference. Connect grounds first, then connect the control signal. Do not assume two separate bench-supply outputs share a ground unless you have verified it.

Board Reference

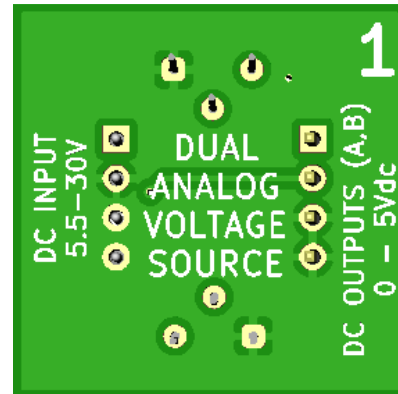
The following pages provide operating instructions, limits, and theory for each module. Connector labels printed on the boards are authoritative. Remove power before changing any connection or jumper.

Board 1 Dual Output Analog Control Voltage Source

Provides two independent adjustable control voltages for analog-dimmable LED drivers. Each output is set with a single-turn trimmer.



Component side



Connection side

Setup

1. Connect a 5.5 V to 25 V DC supply to the power input.
2. Connect output A or B to an analog-control input.
3. Measure the output before connecting a driver, then adjust from nearly 0 V toward the desired level.

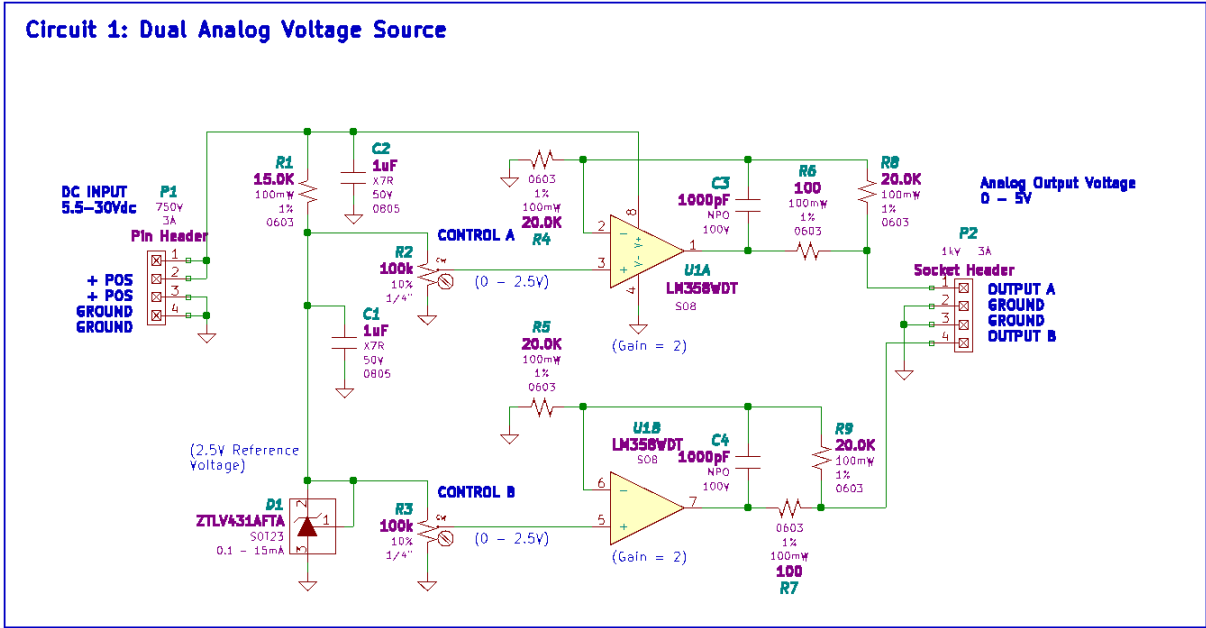
Specifications

Parameter	Specification	Notes
Supply voltage	5.5 V to 25 V	Output range is reduced when operated from 5 V
Output range	<10 mV to >4.95 V	Load resistance greater than 1 k Ω
No-load input current	8 mA maximum	At 25 V
Protection	Short-circuit tolerant	LM358 provides internal current and thermal protection

How It Works

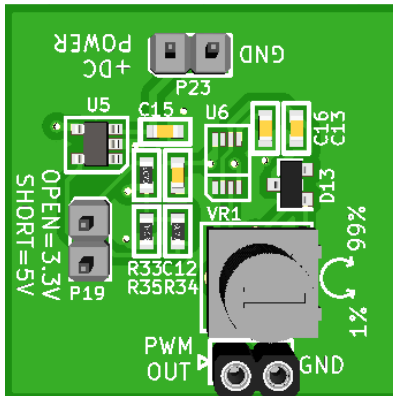
A 2.5 V reference feeds two gain-of-two amplifiers. The potentiometers divide the reference voltage, allowing each output to be adjusted independently over approximately 0 V to 5 V.

Board 1 Schematic

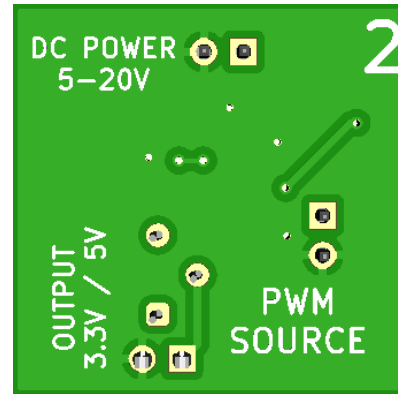


Board 2 PWM Generator

Generates an approximately 100 Hz square wave with an adjustable duty cycle. A jumper selects a nominal 3.3 V or 5 V output level.



Component side



Connection side

Setup

1. Connect a 5 V to 20 V DC supply. Do not exceed 20 V.
2. Set the output-voltage jumper for the driver being controlled.
3. Connect PWM OUT and GND to the target driver. Rotate the trimmer to vary duty cycle.

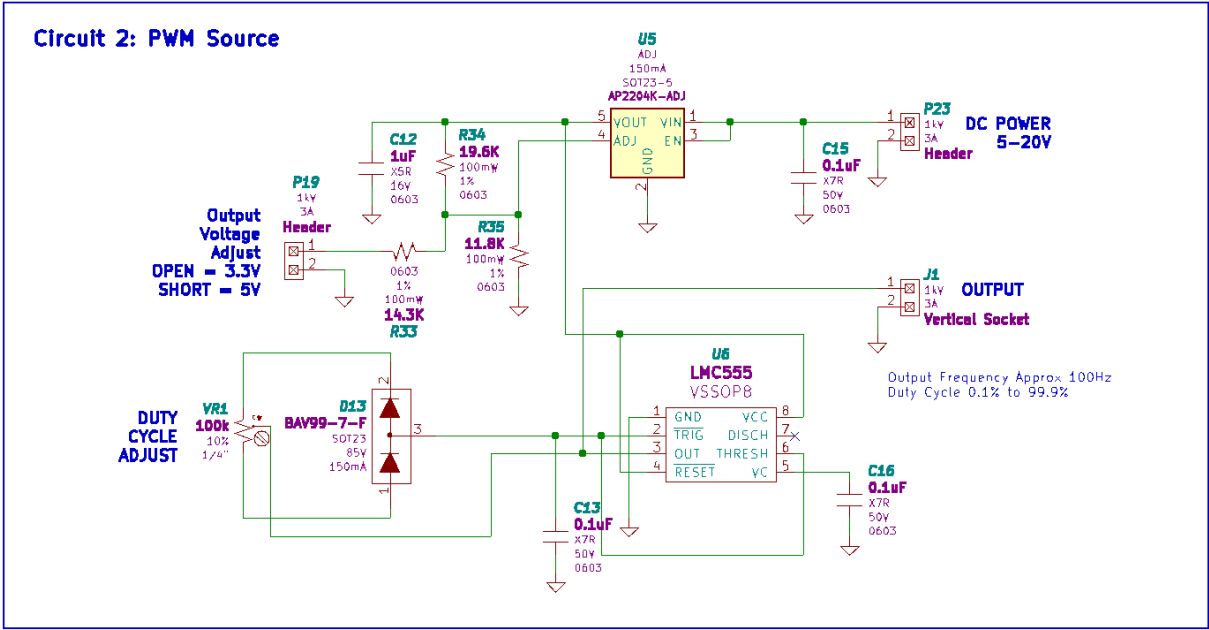
Specifications

Parameter	Specification	Notes
Supply voltage	5 V to 20 V	Do not use the older 15-30 V example value
No-load supply current	<2 mA	Typical operating condition
Output level	3.3 V or 5.0 V	5 V requires supply above 5.5 V
Duty-cycle range	<1% to >99%	Approximately
Frequency	100 Hz typical	Suitable for visible-light experiments

How It Works

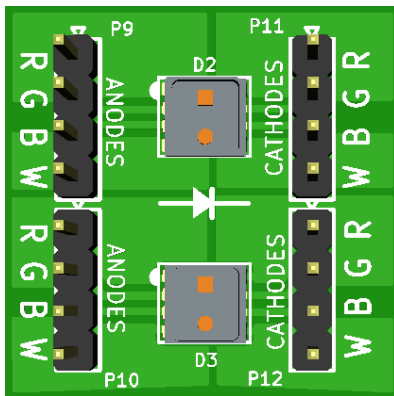
An LMC555 oscillator charges and discharges a timing capacitor through opposite portions of the trimmer. Moving the wiper changes the ratio of charge time to discharge time and therefore the PWM duty cycle.

Board 2 Schematic

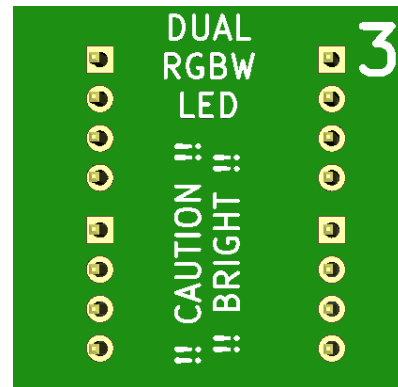


Board 3 Dual RGBW LED Load

Provides two four-color LED packages. Separate anode and cathode connections make it possible to mix red, green, blue, and white channels and compare driver circuits.



Component side



Connection side

Setup

1. Choose one LED package and one or more colors.
2. Connect each selected color only through a constant-current driver.
3. Begin at low current and avoid looking directly into an illuminated LED.

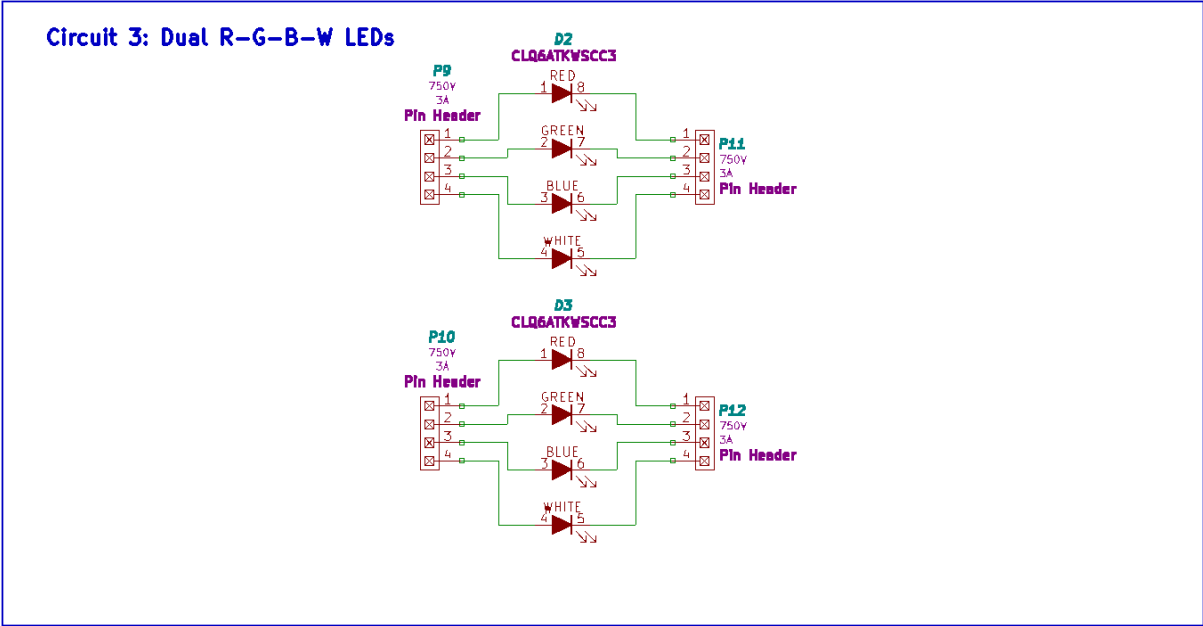
Specifications

Parameter	Specification	Notes
Maximum continuous current	180 mA per die	Use adequate current regulation
Maximum peak current	400 mA per die	Do not treat as a continuous rating
Forward voltage at 100 mA	Red 2.6 V typical	Green and blue 3.8 V; white 3.6 V

How It Works

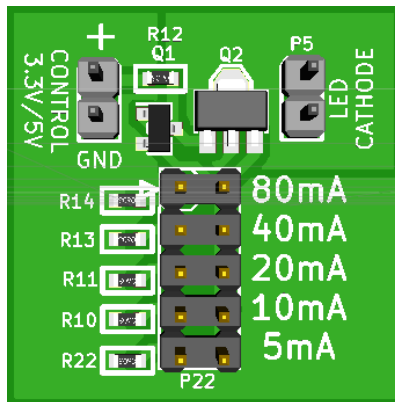
Each package contains four independent LED dice. Their separate connections let different driver boards control the colors independently for additive color-mixing experiments.

Board 3 Schematic

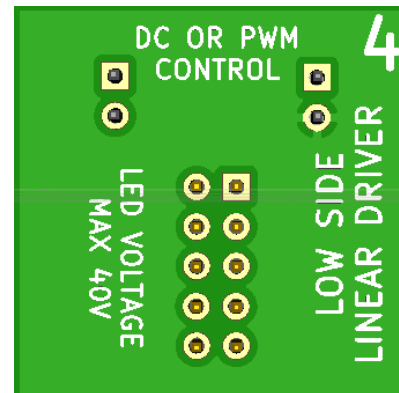


Board 4 Low Side Linear Driver with PWM Dimming

A low-side constant-current sink with jumper-selected current and a digital ENABLE/PWM input.



Component side



Connection side

Setup

1. Install jumpers whose labeled currents add to the desired set point.
2. Connect the LED cathode to the driver and the LED anode to the positive supply.
3. Drive ENABLE/PWM above 2 V to turn on, or apply PWM from Board 2.

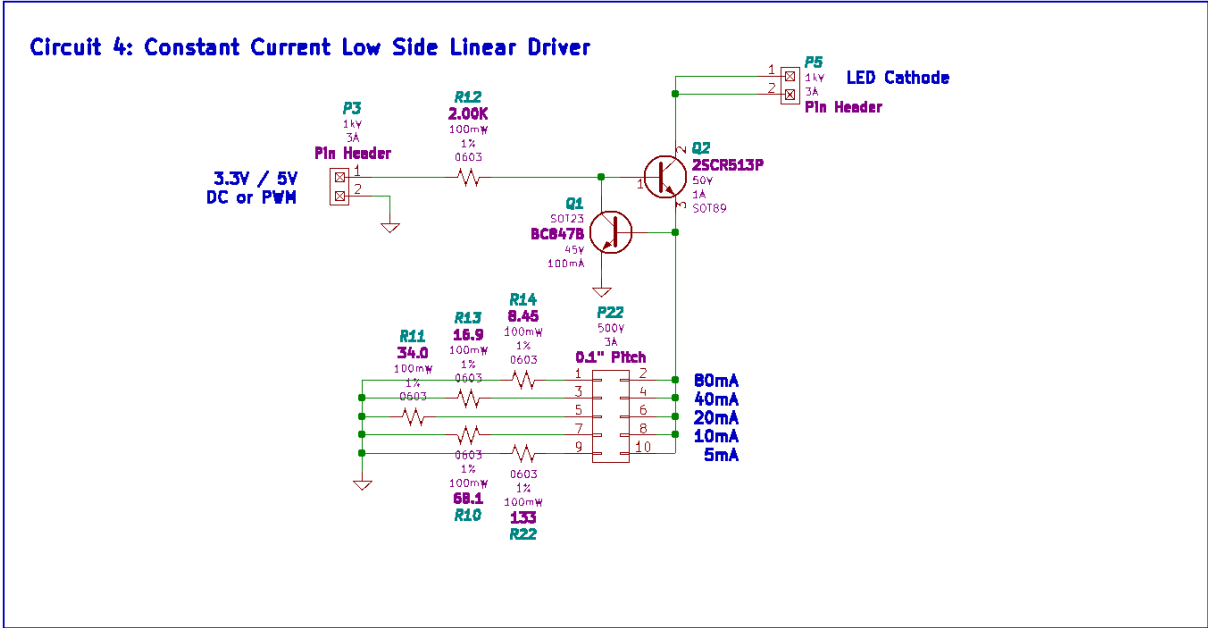
Specifications

Parameter	Specification	Notes
Current selection	5 mA to 155 mA	5 mA increments by adding jumper values
Enable high level	>2 V	Turns the LED current on
Dropout voltage	1.0 V typical	At 155 mA
Output voltage capability	Up to 60 V	Voltage across the driver, within device ratings

How It Works

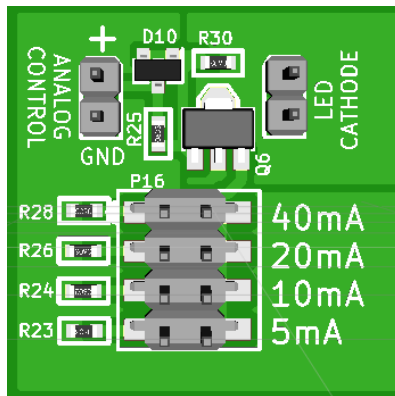
Q1 carries the LED current. When the voltage across the selected sense resistance reaches about 0.65 V, Q2 limits additional drive to Q1. During PWM high time, current is regulated at the jumper-selected value; average LED current equals set current multiplied by duty cycle.

Board 4 Schematic

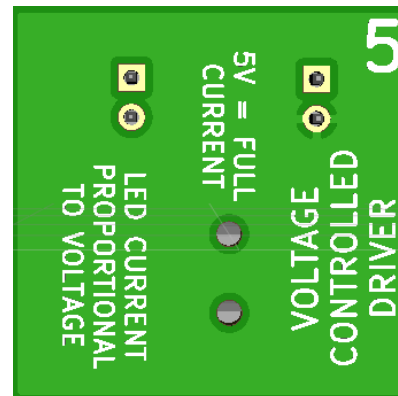


Board 5 Low Side Linear Driver with Analog Dimming

A simple low-side voltage-controlled current sink. Jumpers establish the full-scale range, and the analog input varies LED current continuously.



Component side



Connection side

Setup

1. Select a maximum current by adding the enabled jumper values.
2. Connect the LED string above the driver.
3. Apply 0 V to 5 V from Board 1 and increase slowly while observing current and temperature.

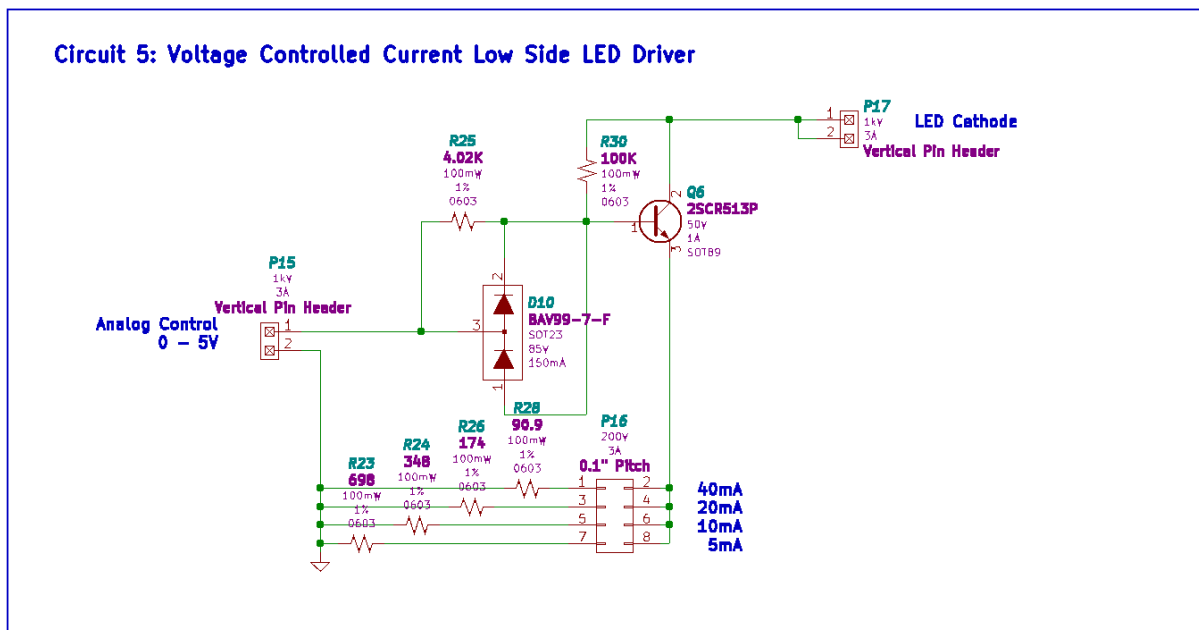
Specifications

Parameter	Specification	Notes
Full-scale current	5 mA to 75 mA	5 mA increments; specified at 5 V control
Dropout voltage	0.5 V typical at 1 V control	Approximately 4.5 V at 5 V control
Output voltage capability	Up to 60 V	Voltage across the driver, within device ratings

How It Works

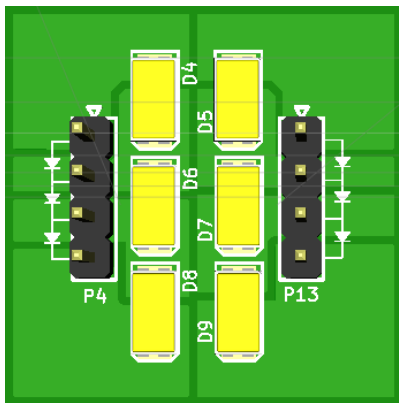
The error amplifier drives the pass transistor so that sense voltage follows the analog-control voltage. Selected resistance determines the current scale. This topology is intentionally simple, but its dropout voltage rises with the control voltage.

Board 5 Schematic

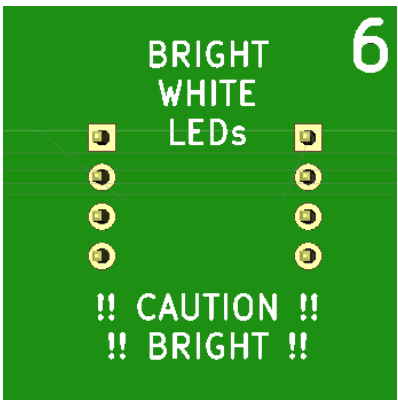


Board 6 Two Strings of High Brightness White LEDs

Provides two independent strings of three series-connected white LEDs for linear and switching-driver experiments.



Component side



Connection side

Setup

- 1. Connect only through a constant-current driver.
- 2. Start below 50 mA until wiring and polarity are verified.
- 3. Use the test points to compare string voltage and brightness at different currents.

Specifications

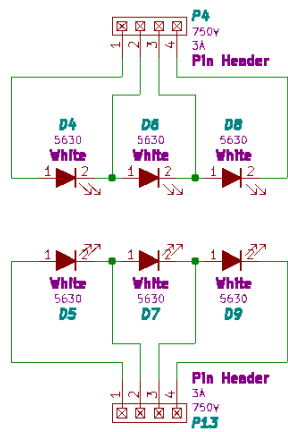
Parameter	Specification	Notes
Maximum continuous current	240 mA per LED	Observe temperature carefully
Forward voltage	2.8 V typical per LED	3.0 V maximum at 65 mA
Light output	38 lm typical per LED	At 65 mA, device-dependent

How It Works

Three series LEDs require roughly three times the forward voltage of one LED while carrying the same current. The two strings allow comparison, parallel experiments, or use with separate channels.

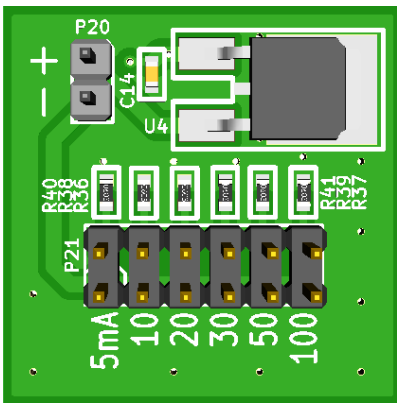
Board 6 Schematic

Circuit 6: White LEDs

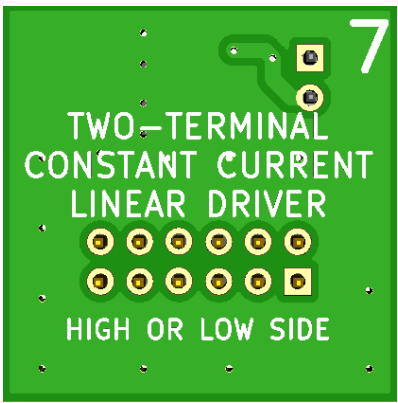


Board 7 Two Terminal Constant Current Linear Driver

A two-terminal LM317-based current regulator that can be inserted on the high side, low side, or anywhere in series with an LED string.



Component side



Connection side

Setup

1. Select current by installing the desired combination of jumpers.
2. Place the board in series with the LED string and supply.
3. Verify that the supply exceeds total LED voltage by at least the required dropout voltage.

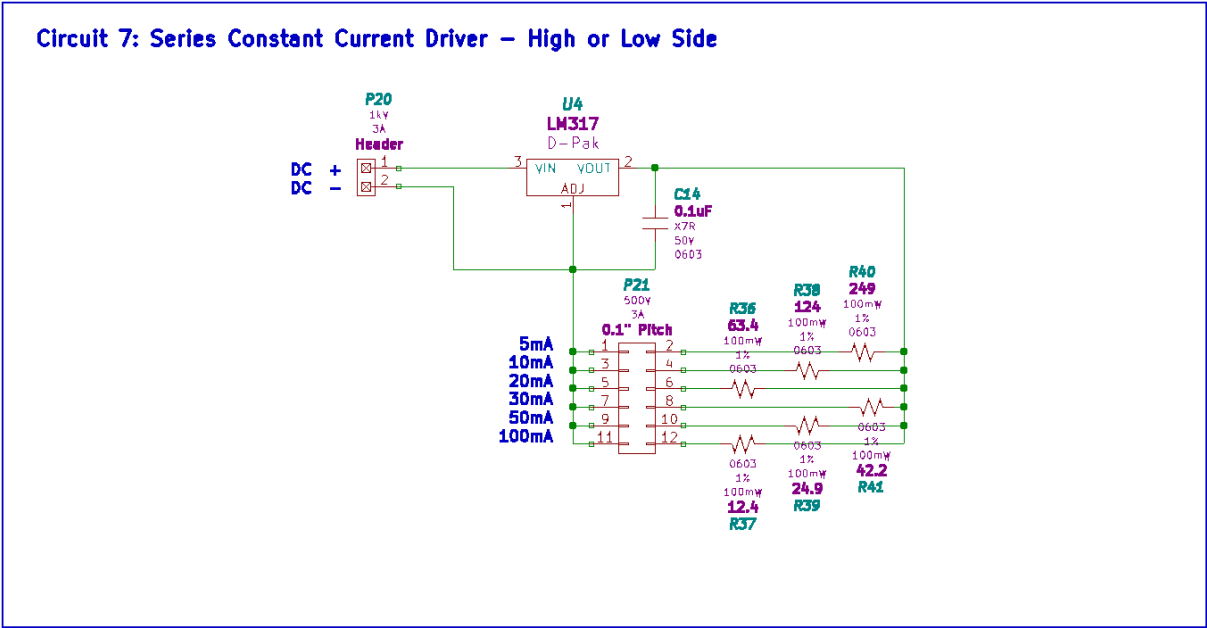
Specifications

Parameter	Specification	Notes
Current selection	5 mA to 215 mA	5 mA increments
Operating headroom	3 V to 32 V	Supply voltage minus total LED voltage
Dropout voltage	2 V maximum at 5 mA	2.5 V maximum at 215 mA

How It Works

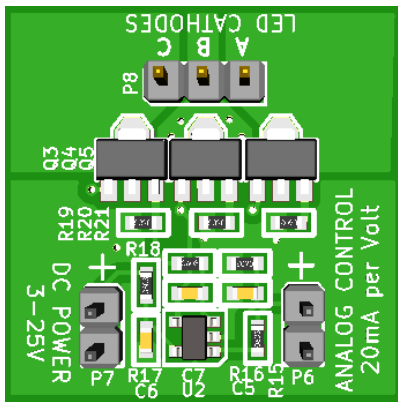
The LM317 regulates approximately 1.25 V across the jumper-selected external resistance. The same current flows through the regulator, resistor, and LED string, so LED current is approximately 1.25 V divided by the selected resistance.

Board 7 Schematic

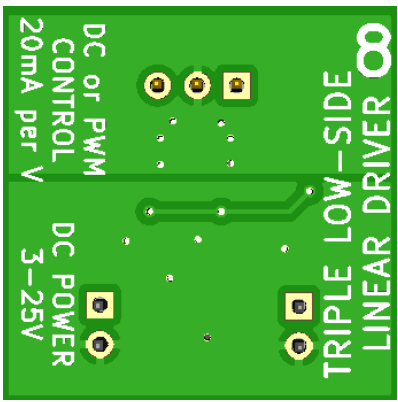


Board 8 Triple Output Linear Driver with Analog or PWM Control

Drives as many as three LED strings at the same current. Its low-pass input filter converts analog voltage or PWM average value into a proportional DC current command.



Component side



Connection side

Setup

1. Connect one to three LED cathodes to the three outputs.
2. Connect Board 1 for direct analog control, or Board 2 for filtered PWM control.
3. Increase control gradually; each output produces approximately 20 mA per volt.

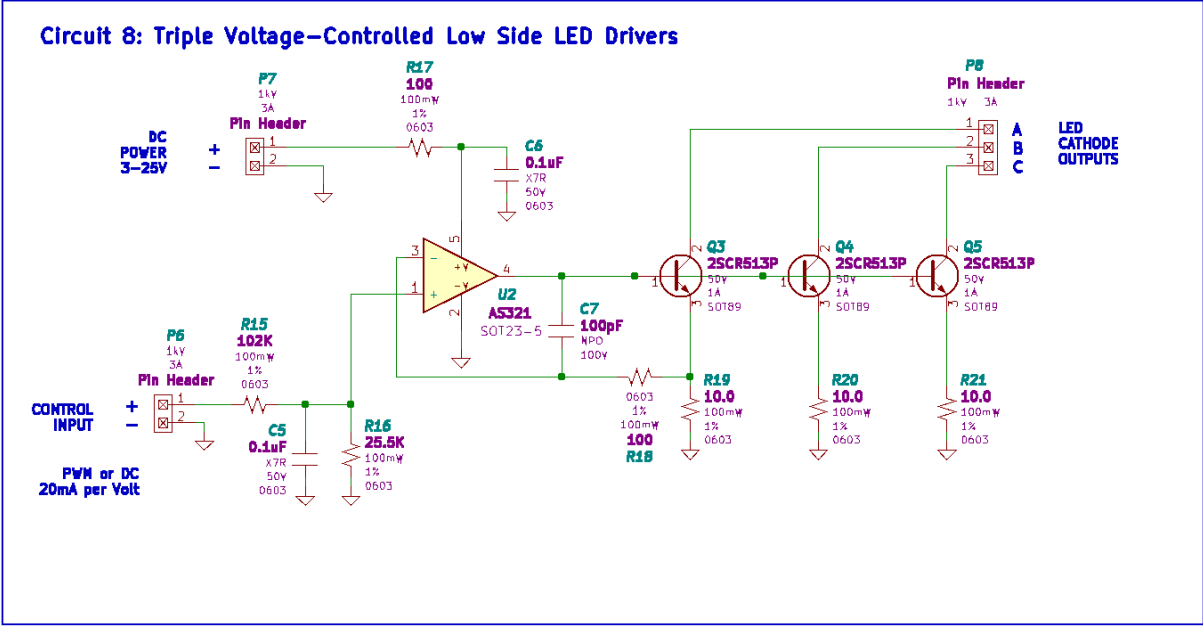
Specifications

Parameter	Specification	Notes
Outputs	3 matched low-side sinks	Each output may be used independently
Maximum output current	100 mA per output	At 5 V control
Transfer function	20 mA/V typical	For control below 5 V
Collector voltage	50 V maximum	Observe overall board and transistor ratings

How It Works

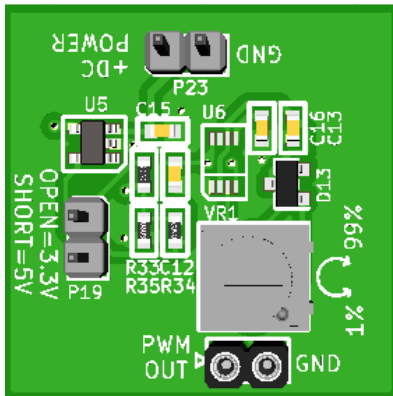
The input filter averages a PWM waveform into a DC voltage. An error amplifier and current-sense network command three output transistors to track that average voltage. With a 5 V PWM waveform at 20% duty cycle, the average is about 1 V and each active output is approximately 20 mA.

Board 8 Schematic

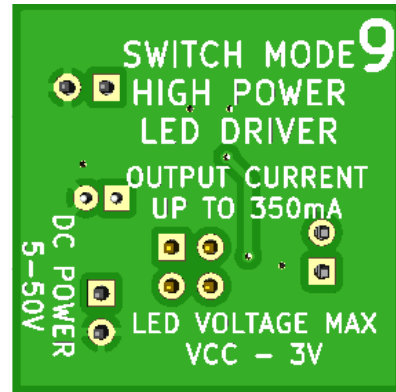


Board 9 High Efficiency Switch Mode LED Driver

A ZXLD1356 buck LED driver for efficient constant-current operation of higher-voltage LED strings, with digital PWM dimming.



Component side



Connection side

Setup

1. Set the current jumpers before applying power. Values add to a maximum of 350 mA.
2. Use an input voltage at least about 2 V above the LED-string voltage.
3. Apply a 3.3 V or 5 V PWM signal. Never exceed 7 V on the PWM input.

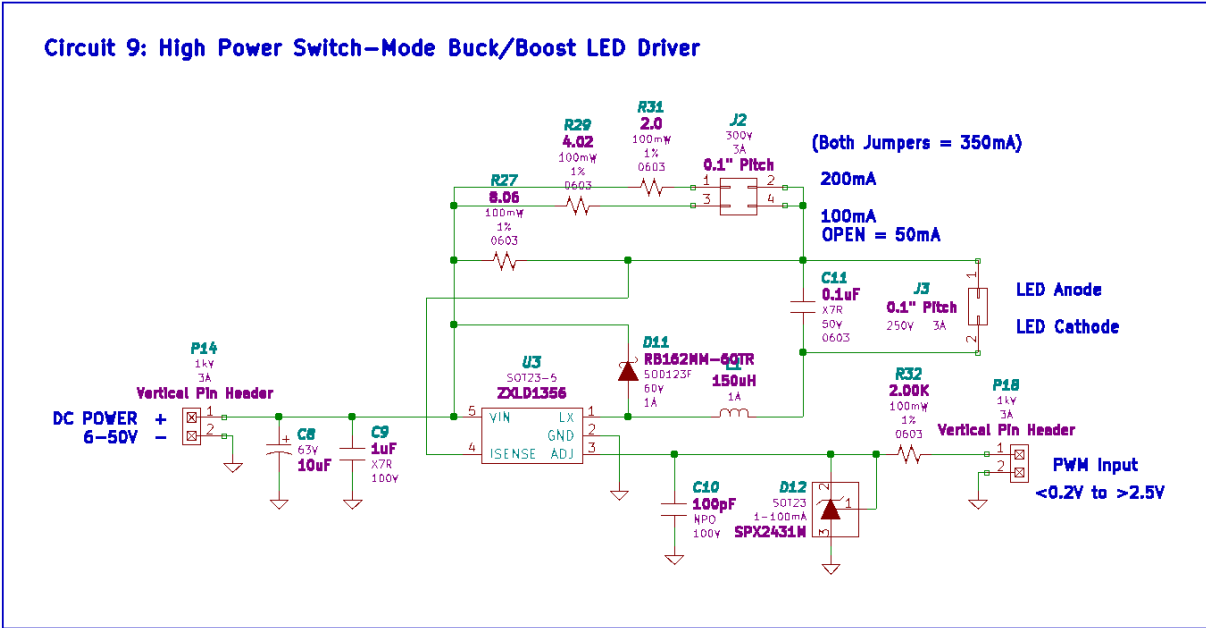
Specifications

Parameter	Specification	Notes
Supply voltage	6 V to 50 V	Buck topology requires input above LED voltage
Output current	Up to 350 mA	Jumper-selected; values are additive
Dropout voltage	2 V maximum	PWM high
PWM low/high	<0.3 V / >2.5 V	7 V absolute maximum
Efficiency	Typically 78% to 95%	Depends strongly on input and LED voltage

How It Works

The ZXLD1356 switches current through an inductor and senses LED current through a low-value resistor. Energy conversion is substantially more efficient than burning excess voltage in a linear pass element. PWM disables and enables switching without changing the programmed on-state current.

Board 9 Schematic



Using a Microcontroller

Boards 4 and 9 accept logic-level PWM control and can be driven by many microcontrollers. Board 8 can accept either a DC control voltage or a PWM signal that its input filter averages. Confirm the required logic level and maximum input voltage before connecting any controller.

- Connect the microcontroller ground to the LED-driver ground before connecting the signal.
- Use a PWM frequency in the approximate 100 Hz to 200 Hz range for initial testing. Verify correct operation before exploring other frequencies.
- Never apply more than 7 V to Board 9's PWM input.
- A Raspberry Pi GPIO is 3.3 V logic and is not 5 V tolerant. Do not allow a kit board to drive voltage back into the GPIO.
- The microcontroller supplies only the control signal. Power LEDs from a suitable external DC supply, not from a GPIO pin.

Example Pseudocode

```
set_pwm_frequency(100)
for duty in range(0, 101, 5):
    set_pwm_duty_cycle(duty)
    wait(0.10)
```

Troubleshooting

Symptom	Checks
No light	Remove power. Check supply polarity, current limit, common ground, LED polarity, jumper placement, and control-input level. Measure the LED-string voltage before increasing the supply.
Only one color works	Check the separate anode and cathode pins for each RGBW die. Verify that every active color has its own current-regulated path.
Brightness does not change	Measure the control voltage or PWM waveform at the driver input. Check the Board 2 output-level jumper and confirm that grounds are connected.
Driver becomes hot	Reduce current or reduce the difference between supply voltage and LED voltage. Estimate linear-driver dissipation and allow the board to cool.
Board 9 does not regulate	Confirm that input voltage is above LED-string voltage by approximately 2 V, PWM is above 2.5 V when high, and current-selection jumpers are installed correctly.
Visible flicker	Increase PWM frequency toward 100 Hz or above and confirm that the control waveform is stable.
Bench supply enters current limit	Turn off power and inspect for reversed wiring or a short circuit. Restart at a lower programmed LED current.

Useful Equations

Parameter	Specification	Notes
PWM average current	$I_{AVG} \approx I_{ON} \times \text{duty cycle}$	Approximate; verify by measurement
Linear-driver dissipation	$P_{DRIVER} \approx (V_{SUPPLY} - V_{LED}) \times I_{LED}$	Approximate; verify by measurement
LED string voltage	$V_{STRING} \approx \text{sum of the individual LED forward voltages}$	Approximate; verify by measurement
Board 7 current	$I_{LED} \approx 1.25 \text{ V} \div R_{SET}$	Approximate; verify by measurement
Board 8 current	$I_{LED} \approx 20 \text{ mA/V} \times V_{CONTROL}$	Approximate; verify by measurement

Revision note: This manual reconciles the original board data sheets and application examples. Board 2 is limited to 20 V maximum, Board 9 percentage figures are identified as efficiency, and board names have been standardized throughout.

Application Setup Diagrams

These diagrams show the physical interconnections between modules. They include both board photographs and simplified functional drawings so connections can be checked two ways before power is applied.

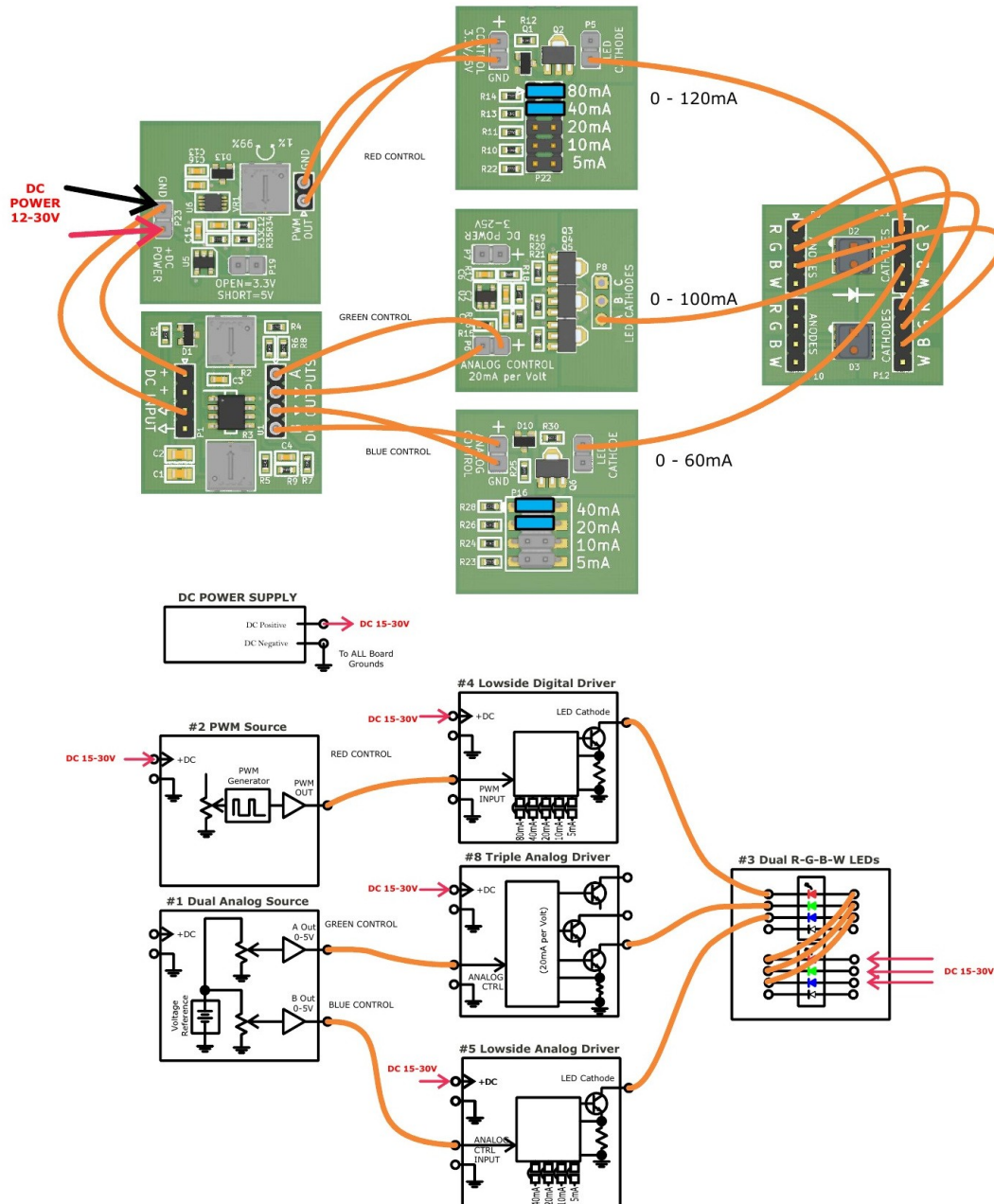
RGB Color Mixing

Uses Boards 1, 2, 3, 4, 5, and 8 to compare PWM, analog, and filtered-PWM dimming while independently controlling red, green, and blue.

Voltage correction: When one supply powers every board, use 15 V to 20 V. The legacy drawing shows values as high as 30 V, but Board 2 is limited to 20 V, while Boards 1 and 8 are limited to 25 V.

Application Example

R - G - B Color Mixing (Uses boards #1, 2, 3, 4, 5, and 8)



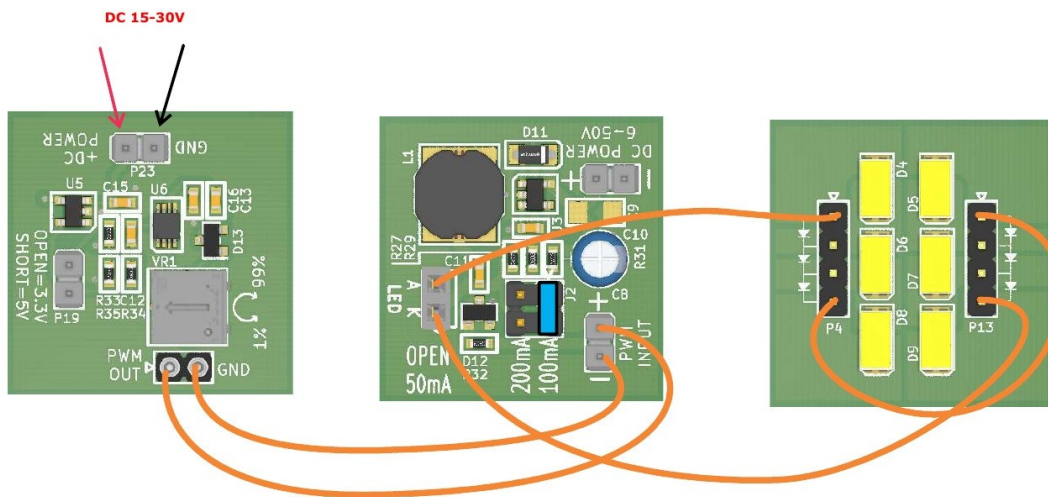
High Efficiency Bright LED Driver with PWM Dimming

Uses Board 2 as the PWM source, Board 9 as the switch-mode constant-current driver, and Board 6 as the white-LED load.

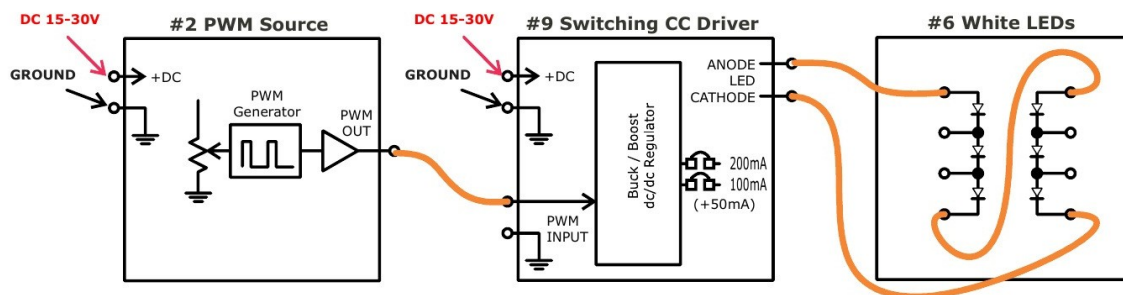
Voltage correction: For a shared supply, use 15 V to 20 V rather than the 15 V to 30 V range printed on the legacy diagram. Board 2 must not exceed 20 V.

Application Example

High Efficiency BRIGHT LED Driver with PWM dimming. (Uses boards #2, 6, and 9)



CAUTION: Do NOT Look Directly at LEDs
Possible Eye Damage



Simple Constant Current Linear Driver

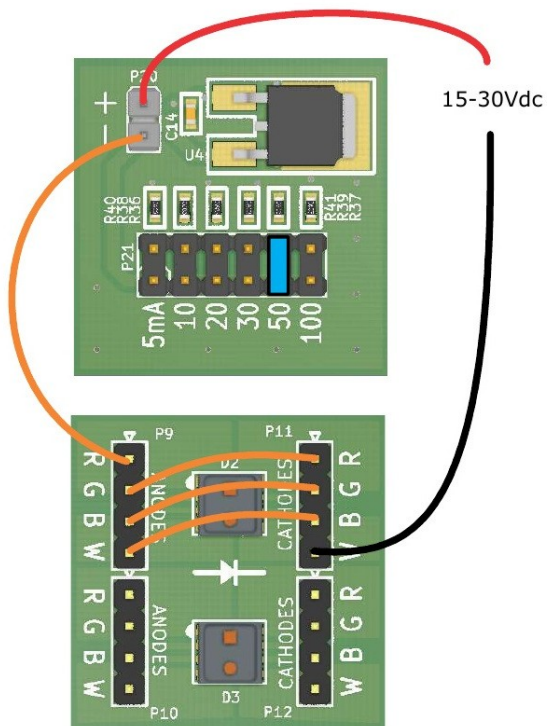
Uses Boards 3 and 7 to demonstrate that the two-terminal constant-current driver can be placed on either the high side or the low side of an LED string.

Application Example

Simple Constant Current Linear Driver (Non-dimmable) (Uses boards #3 and 7)

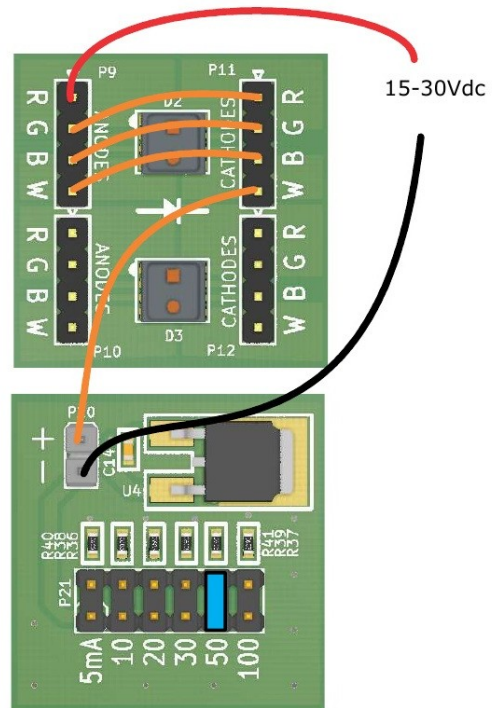
High-Side Connection

50mA Constant-Current LED Driver
High-Side Driver (Driver on the positive side)



Low-Side Connection

50mA Constant-Current LED Driver
Low-Side Driver (Driver on the ground side)



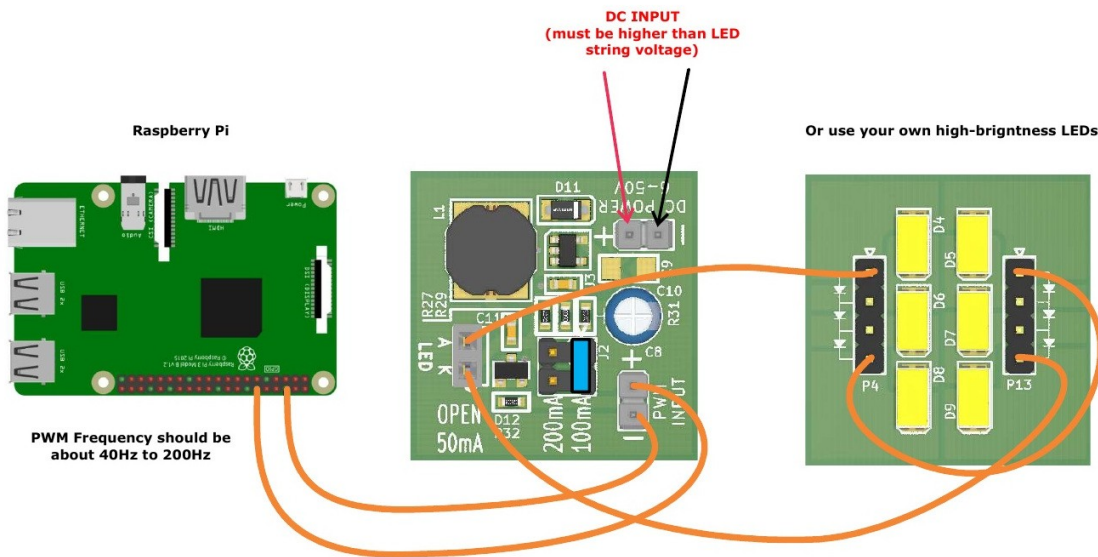
Microcontroller Controlled High Power LEDs

Shows how a Raspberry Pi or other 3.3 V logic source can provide PWM control to Board 9 while the LED power comes from an external supply.

Voltage correction: Connect controller ground to driver ground. Never apply more than 7 V to Board 9 PWM INPUT, and never allow 5 V to feed back into a Raspberry Pi GPIO.

Using boards with microcontrollers like Arduino and Raspberry Pi

Several of these boards can be used with microprocessors to drive high power LEDs. For example, a microprocessor can be used to turn on an LED illuminator being used with a camera or motion sensor. Some of the drivers can drive several series-connected LEDs to generate a very bright light source. For example, Board #9 can easily drive a string of ten high brightness LEDs at 1W each (350mA at 3V), resulting in light output of approximately 1000 lumens. Using a PWM drive to the board allows dimming. See this article for example:
<https://www.mbttechworks.com/projects/raspberry-pi-pwm.html>



End of Manual